Examiner only

10. *(a)* The mass spectrum of a silicon sample showed the presence of the three isotopes ^{28}Si, ^{29}Si and ^{30}Si. The percentage of ^{28}Si present was 92.2 % and the mass spectrum showed that the percentage of ^{29}Si present was twice that of ^{30}Si.

Calculate the relative atomic mass of this sample of silicon. [3]

You **must** show your working.

Relative atomic mass = ...

(b) Rhiannon studied the properties of silicon and found that its structure is similar to that of diamond. In her report she stated that

- the Si–Si–Si bond angle is 109.5°
- silicon is a poor conductor of electricity
- each silicon atom is bonded to four other silicon atoms.

(i) State the name of the shape that has this bond angle. [1]

....

(ii) Explain why solid silicon is a very poor electrical conductor. [1]

....

....

(iii) Explain why the bonding between each atom is covalent. [1]

....

....

....

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(c) Hydrofluoric acid, HF, is supplied as a solution that contains 50% by mass of hydrogen fluoride. The density of this solution is 1.17 g cm^{-3}.

Calculate the concentration of this solution in mol dm^{-3}. [2]

Concentration = ... mol dm^{-3}

(d) (i) Hydrofluoric acid will dissolve silica, SiO_2, to produce hexafluorosilicic acid. This acid contains the SiF_6^{2-} ion.

Use the information from the table below to draw the shape of the SiF_6^{2-} ion, showing the F–Si–F bond angle. Give a reason for your answer. [3]

Ion	Number of bonding electron pairs	Number of lone electron pairs on the central silicon atom
SiF_6^{2-}	6	0

....

....

....

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(ii) Hexafluorosilicic acid, H_2SiF_6, (M_r 144) can be added to drinking water to promote good dental health. When added to water all the fluorine in the acid is available as fluoride ions.

The water in an area of Derbyshire contains 0.15 mg dm^{-3} of fluoride ions.

Calculate how much hexafluorosilicic acid, in mg, should be added to each dm^3 of water to increase the fluoride level to 0.76 mg dm^{-3}. [3]

Mass of hexafluorosilicic acid = ... mg

(e) Solid magnesium hexafluorosilicate hydrolyses rapidly when added to water. One equation for this reaction is as follows.

$$MgSiF_6(s) + 2H_2O(l) \longrightarrow Mg^{2+}(aq) + 6F^-(aq) + SiO_2(s) + 4H^+(aq)$$

In an experiment 2.60 g of magnesium hexafluorosilicate were added to 1.00 dm^3 of water.

Calculate the pH of the resulting mixture. [3]

pH = ...

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